



Centre for
Robotics



Trajectory Tracking via Multiscale Continuous Attractor Networks

Therese Joseph, Tobias Fischer and Michael Milford

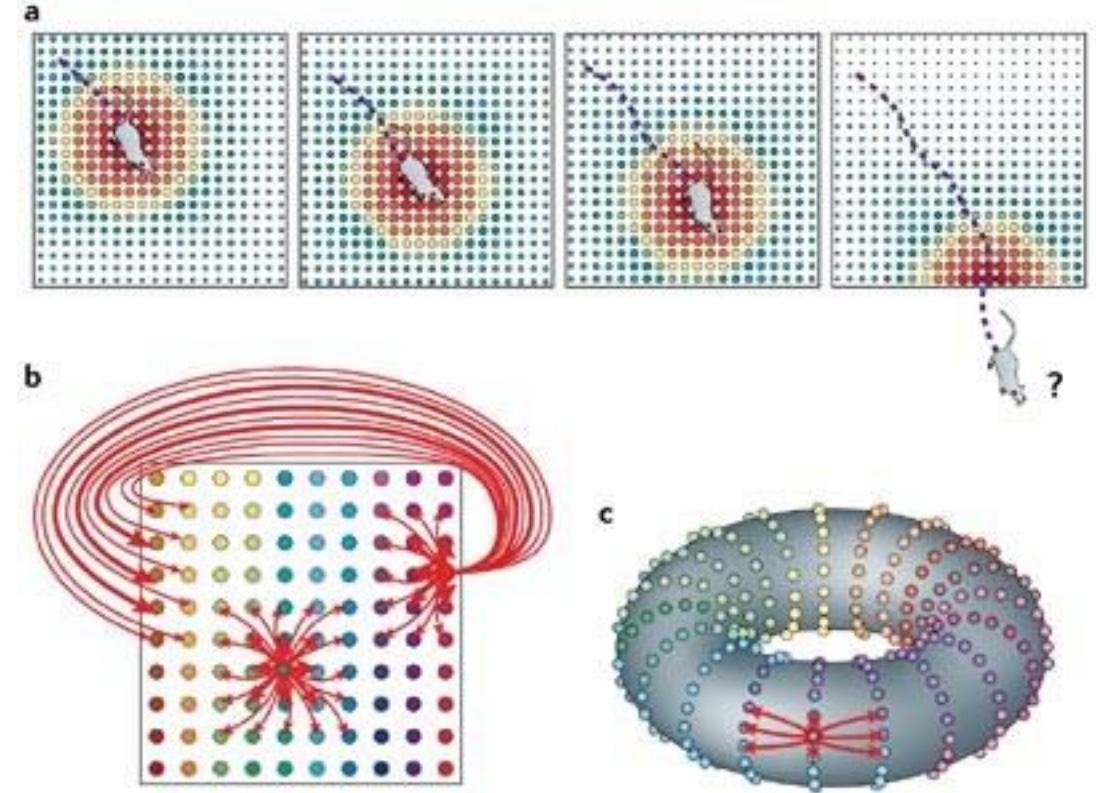
QUT Centre for Robotics

Motivation and Contribution

- Bio-inspired navigation algorithms like RatSLAM and NeuroSLAM are highly handcrafted systems that have difficulty scaling and adapting to a wide range of spatial scales.
- We propose a bio-inspired neural network architecture that combines parallel networks of previous theoretical models to represent multiple spatial scales.
- Our contributions include:
 - Multiscale Continuous Attractor Networks
 - A tuning method using genetic algorithms
 - A city-scale navigation simulator

Attractor Models for Spatial Representation

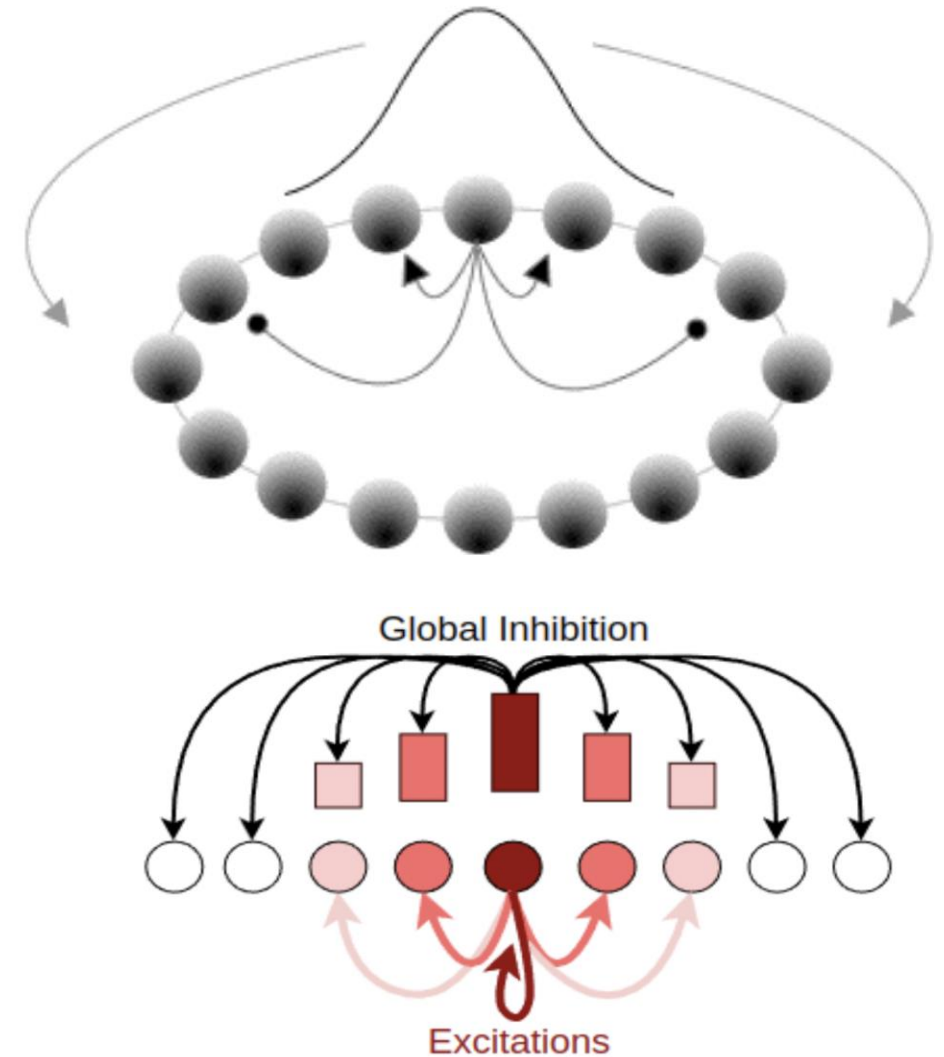
- An attractor network is a recurrently connected set of nodes that settles on to a stable bumps of activity corresponding to a preferred state.
- McNaughton et al. proposed a model that propagates activity bumps to represent movement through space with continuous information about speed and direction.
- To efficiently reuse the attractors, the the edges of the network wraparound forming a toroidal manifold.



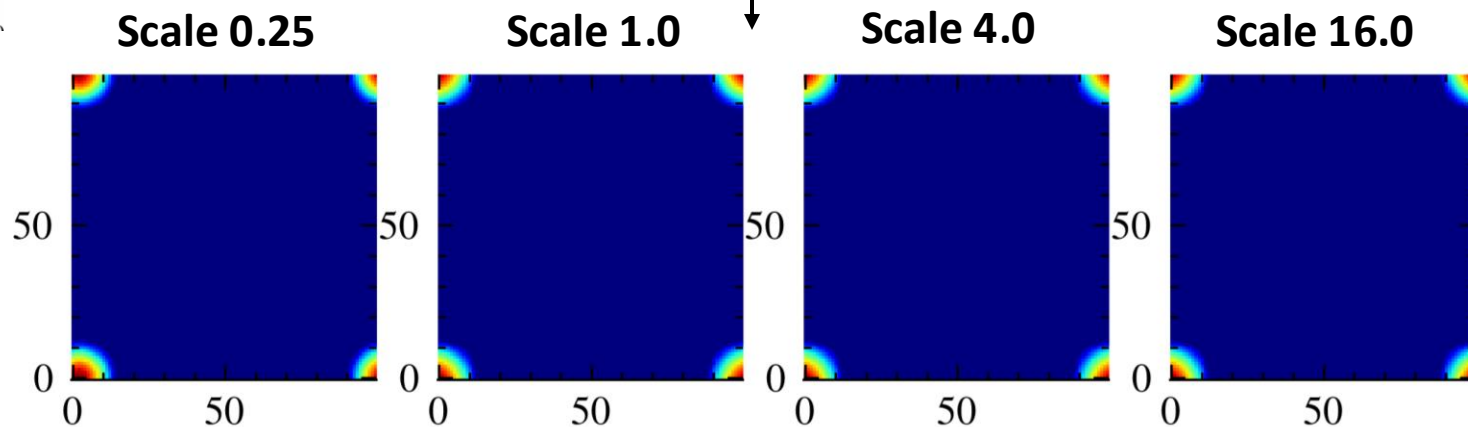
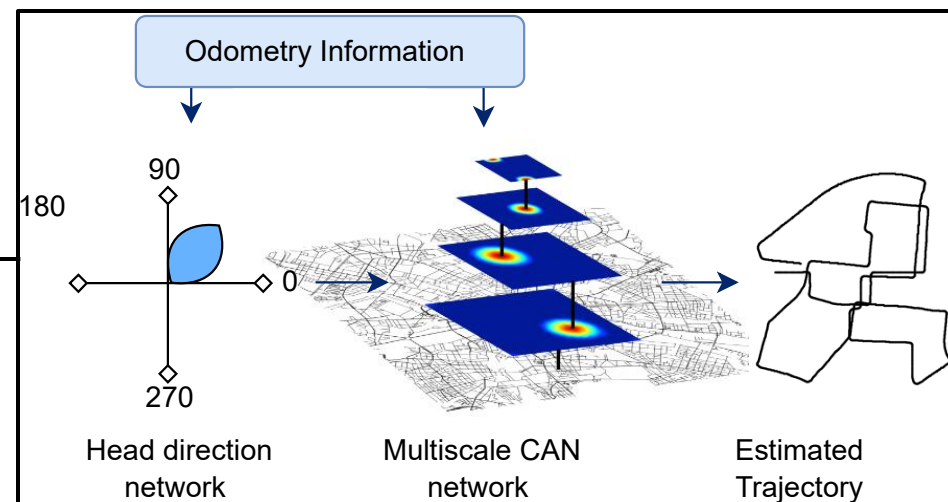
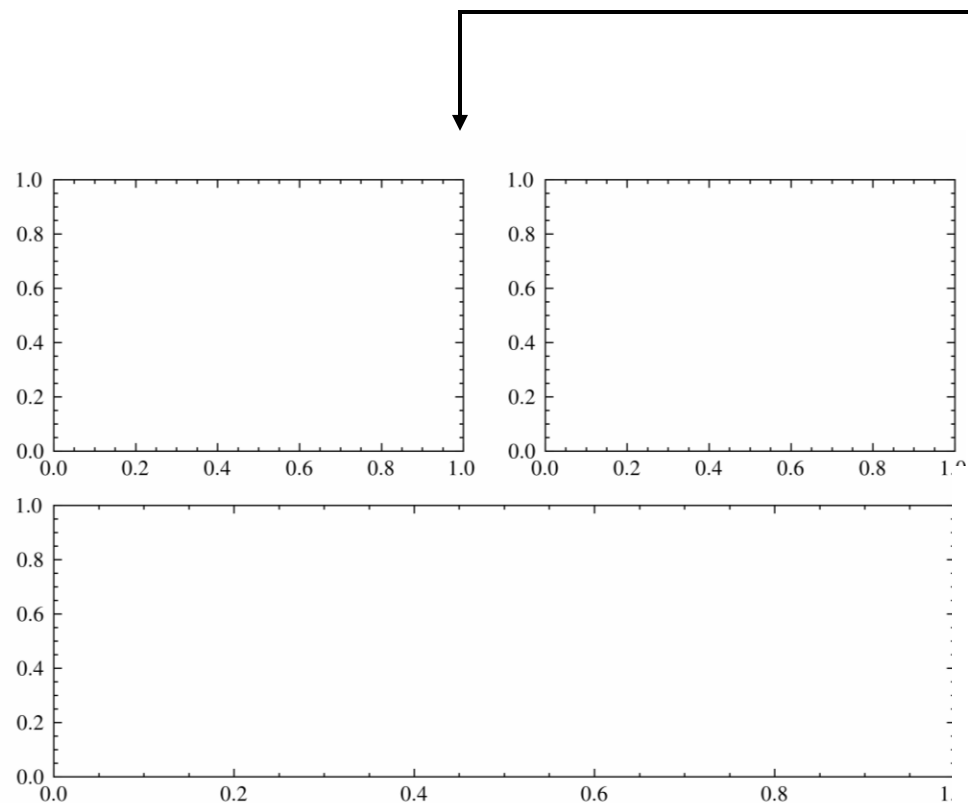
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Attractor Dynamics

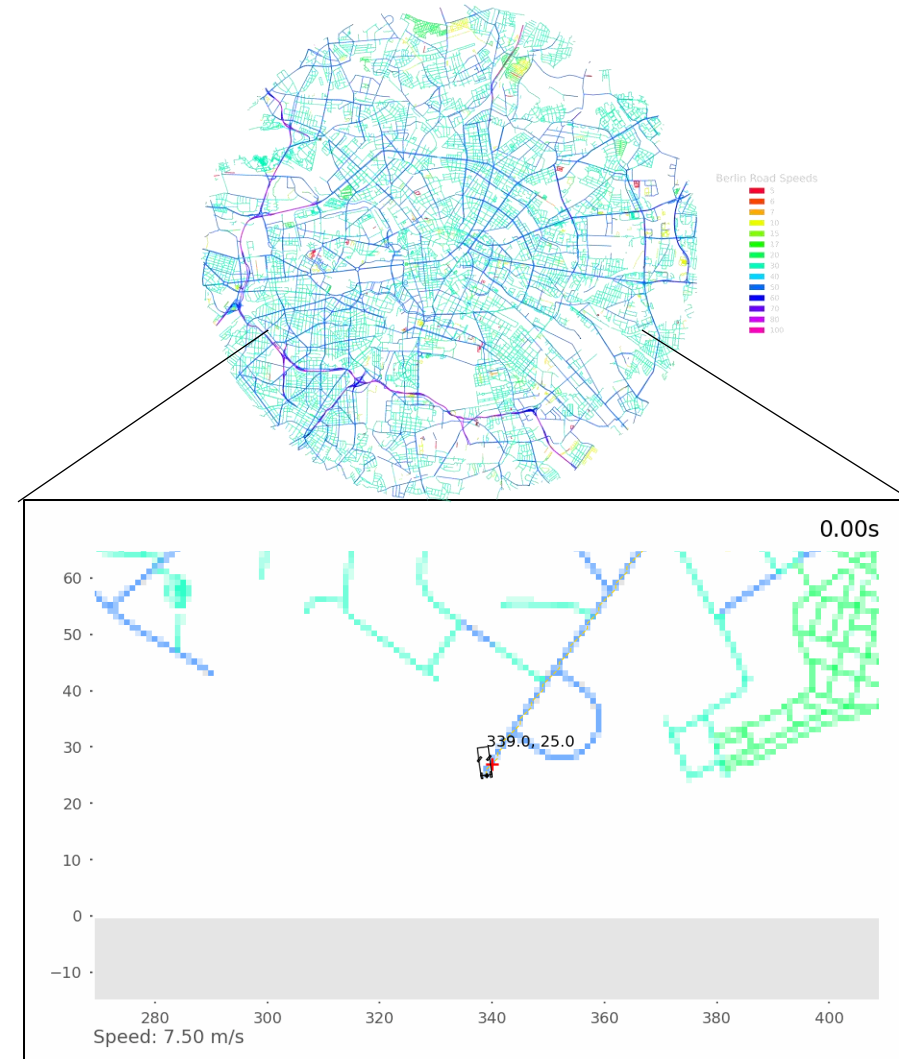
- Attractor dynamics allows inputs to be integrated by the network.
- This is achieved by copying and offsetting existing activity bumps based on the input velocity. The resulting asymmetric bump shifts the activity in the preferred direction.
- The network activity is propagated by exciting neighboring neurons of all active neurons and explosive growth is prevented by inhibiting and normalising the activity after each update.



System Architecture



City-scale navigation simulator

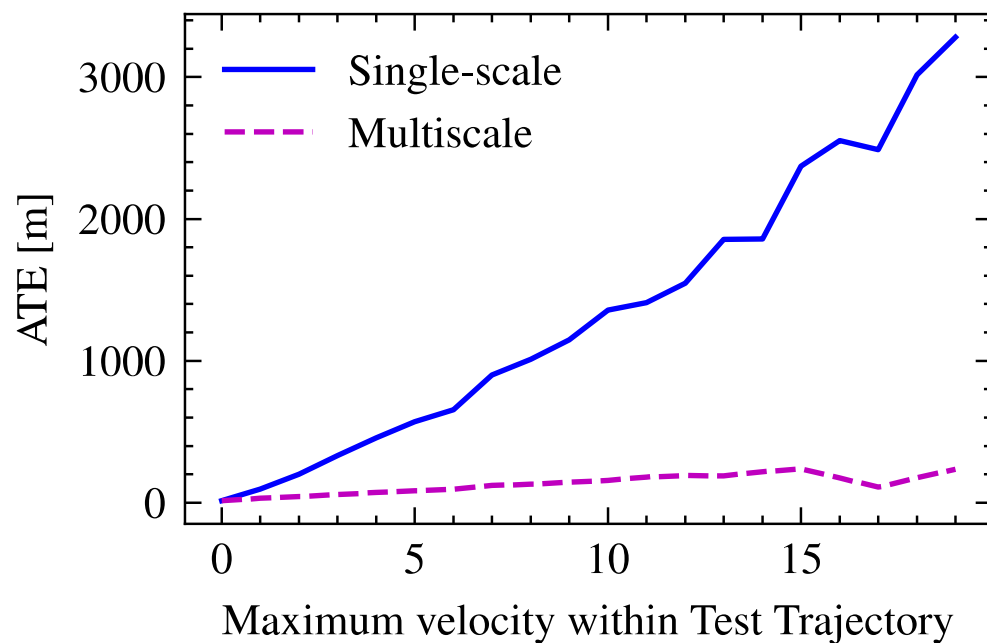


Results

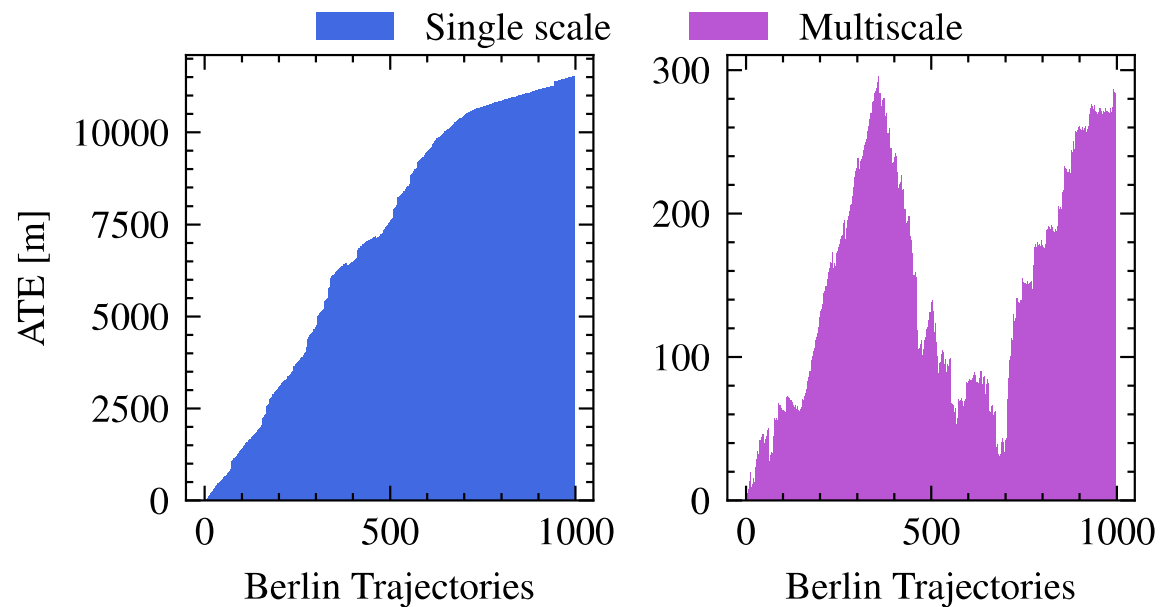
AVERAGE ATE ERROR		
Dataset	Single-scale	Multiscale
Tokyo (CSN simulation)	1.093 ± 0.110	0.068 ± 0.010
New York (CSN simulation)	0.893 ± 0.137	0.070 ± 0.028
Brisbane (CSN simulation)	0.934 ± 0.102	0.070 ± 0.019
Berlin (CSN simulation)	0.896 ± 0.166	0.046 ± 0.021
Kitti (odometry)	0.136 ± 0.138	0.041 ± 0.026

Performance

Network Performance over Large Velocity Ranges

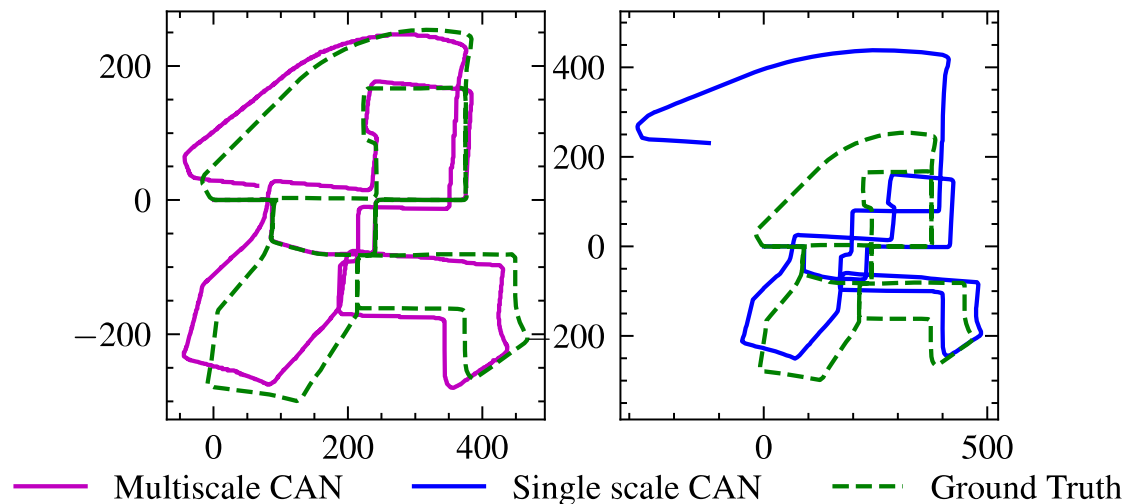


ATE Error Over Time

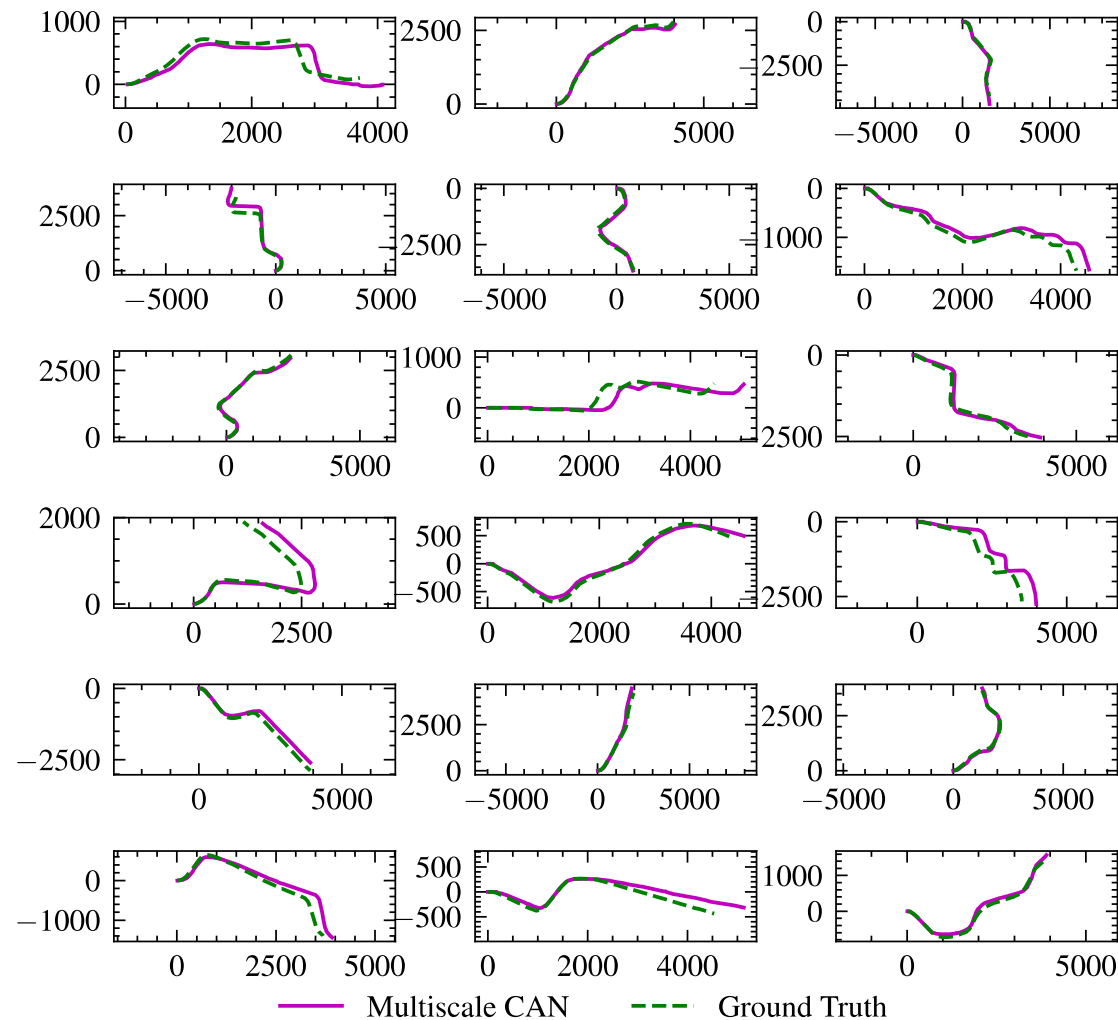


Trajectory Tracking

Multiscale vs. Single Scale Kitti Odometry Path



Tracking Simulated Trajectories through Berlin





Summary

Trajectory Tracking via Multiscale Continuous Attractor Networks

Therese Joseph, Tobias Fischer and Michael Milford

- Has significant performance over the previous single scale systems.
- Capable of accurately encoding a wide range of inputs, enabling trajectory tracking over large velocity ranges
- Utilises automated tuning using genetic algorithms
- Contribute a new flexible city-scale navigation simulator that adapts to any street network, enabling high throughput experimentation.



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